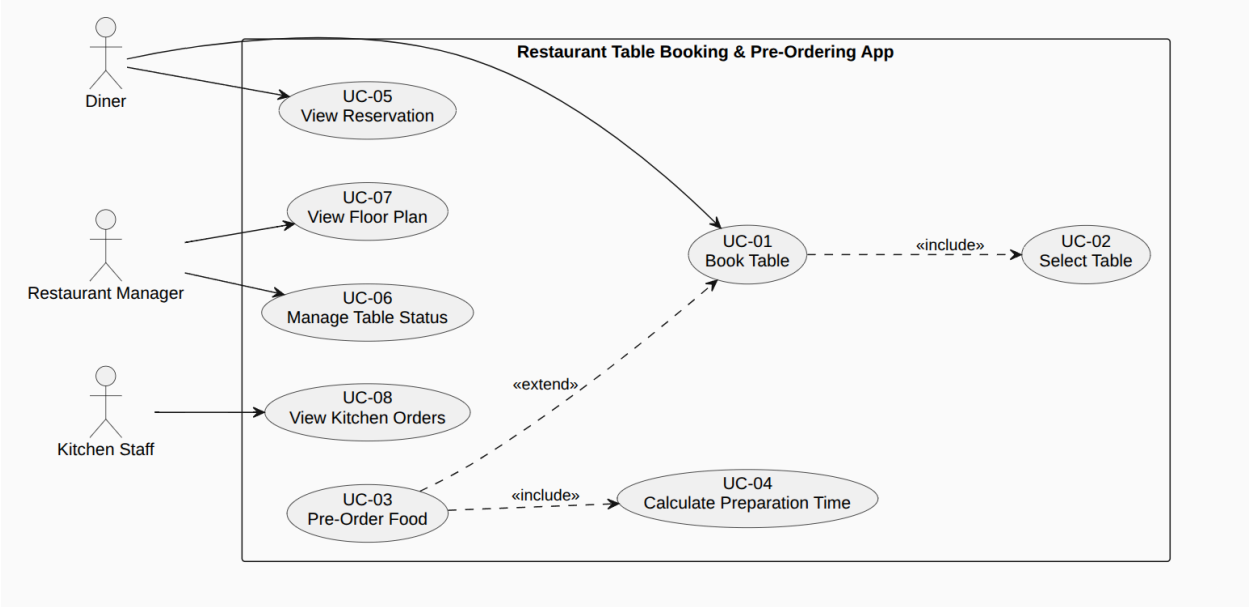
SE-LAB 1

PES2UG24CS102
AYUSHMAN DEY

UML DIAGRAM:-



I added a third actor, Kitchen Staff, to satisfy the requirement of having at least three actors in the use case diagram. I assigned UC-08, View Kitchen Orders, to the Kitchen Staff, since they need to view the orders placed through the pre-ordering process. The Diner and Restaurant Manager keep their original roles and interactions with the system.

REQUIREMENTS TABLE:-

Restaurant Table Booking & Pre-Ordering App

Problem Statement #36 | Retail, E-Commerce & Finance | Actors: Diner, Restaurant Manager

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	Pass: Table reserved and kitchen pre-order ticket queued. Fail: Overbooking of table during overlapping dining window.	Core booking + pre-order functionality (given).	
FR-002	Functional	The system shall allow a diner to browse the restaurant menu and add items to a pre-order linked to their active table reservation.	High	Pass: Selected items appear in the pre-order summary and are attached to the correct reservation ID. Fail: Item added without a valid linked reservation.	Enables the optional pre-ordering flow (extends Book Table).	
FR-003	Functional	The system shall calculate an estimated food preparation time for a pre-order based on selected items and current kitchen queue load.	Medium	Pass: Estimated prep time is displayed within 3 seconds of order submission and is within +/-5 minutes of actual kitchen completion time in test runs. Fail: No estimate shown or estimate off by more than 10 minutes.	Lets kitchen and diner coordinate arrival with food readiness.	
FR-004	Functional	The system shall allow a diner to view their reservation details, including table, date/time, party size, and pre-order status.	Medium	Pass: Reservation screen shows current table, time slot, and pre-order status (Pending/Confirmed/In Prep). Fail: Any of these fields missing or stale by more than 1 minute.	Gives diners confidence and reduces support queries before arrival.	
FR-005	Functional	The system shall allow a restaurant manager to update a table's status (Reserved, Seated, Billed) from the floor plan view.	High	Pass: Status change is saved and reflected on the manager's floor plan within 1 second. Fail: Status update fails silently or does not persist.	Keeps the live floor plan accurate for staff decision-making.	

NFR-001	Nonfunctional (Performance & Security)	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Prevents staff from double-seating a table due to stale terminal data (given).	
NFR-002	Nonfunctional (Security)	The system shall encrypt all diner personal data (name, contact number, order history) using AES-256 both at rest and in transit (TLS 1.2+).	High	Pass: Security audit confirms AES-256 at rest and TLS 1.2+ for all diner data endpoints; no plaintext personal data found in storage or network capture. Fail: Any diner PII found unencrypted at rest or in transit.	Protects diner PII (contact info, dining habits) from breach given restaurant apps store repeat-customer data.	

Use-Case Flow Specification

Restaurant Table Booking & Pre-Ordering App (Problem Statement #36)

Use Case: Book Table (UC-01)

Primary Actor: Diner

Includes: UC-02 Select Table | Extended by: UC-03 Pre-Order Food (optional)

Preconditions:

- Diner has the app open and has specified a valid restaurant location.
- The restaurant's live floor plan data (table availability) is up to date.
- At least one table can accommodate the diner's requested party size for the requested time window.

Postconditions:

- On success: the selected table is reserved for the diner for the requested time window, a confirmation is generated, and (if a pre-order was placed) a kitchen pre-order ticket is queued and linked to the reservation.

- On failure: no table is reserved and the diner is returned to table selection with updated availability.

Main Success Scenario:

1. Diner selects "Book a Table" from the app home screen.
2. System prompts the diner for date, time, and party size.
3. Diner enters the requested date, time, and party size.
4. System displays the interactive 2D floor plan with tables matching the criteria highlighted as available (UC-02: Select Table).
5. Diner selects an open table from the floor plan.
6. System asks the diner whether they would like to pre-order food for this reservation.
7. If the diner opts to pre-order, the system launches the pre-order flow (UC-03: Pre-Order Food), where the diner selects menu items; the system calculates an estimated preparation time (UC-04) and links the order to the reservation.
8. Diner reviews and confirms the booking (with or without a pre-order).
9. System reserves the table, generates a confirmation with a reservation ID, and displays it to the diner.
10. Use case ends successfully.

Alternate Flow :- 5a. Table No Longer Available (Booking Conflict):

a) 5a1. Between the diner loading the floor plan and confirming their selection, another diner reserves the same table for an overlapping time window.

b) 5a2. System detects the conflict at selection time and displays a "Table No Longer Available" message.

c) 5a3. System refreshes the floor plan with current availability and prompts the diner to choose a different table.

d) 5a4. Diner selects a different available table and the flow resumes at step 5.

e) 5a5. If no tables are available for the requested time, the system suggests the nearest available time slots and the diner may retry from step 3.